International Islamic University Chittagong

Department of Computer Science and Engineering
B. Sc. in CSE
Final Term Examination, Spring-2026

Course Code: CSE 3631 **Course Title:** Operating System

Time: 2 hours 30 minutes **Full Marks:** 50

[Answer all the questions. Figures in the right-hand margin indicate course outcomes and marks. Separate answer script must be used for Group A and Group B]

	Group A	CLO	BT	Marks
1.a	A system has three resource types A, B, C and Five processes (P0–P4) have the following Allocation and Maximum matrices: Allocation: P0(1,1,2), P1(2,1,2), P2(4,0,1), P3(0,2,0), P4(1,1,2) Maximum: P0(4,3,3), P1(3,2,2), P2(9,0,2), P3(7,5,3), P4(11,2,3) Available: (2,1,0) Determine: (i) the total amount of resources of each type, (ii) the content of the matrix need, (iii) if the system is "safe" state using the Banker's Algorithm and also safe sequence of the system. Allocation = Current allocation Maximum = Maximum required allocation Available = Current available resources	2	C5	6
1.b	Why Resource Allocation Graph is used? Suppose the system has only one process. Is there any possibility to occur deadlock in that system? Justify your answer.	2	C2	4
	Or			
1.a	A system has four processes (P1–P4) and three resource types (R1, R2, R3), each with a single instance. The current allocation is: R1 → P1, R2 → P2, R3 → P3. The requests are: P1 → R2, P2 → R3, P3 → R1, while P4 is not holding or requesting any resource. Draw the Resource Allocation Graph (RAG) and determine whether the system is in a deadlock. Justify your answer.	2	C5	6
1.b	Explain the role of the four Coffman conditions in deadlock. Illustrate why deadlock cannot occur if any one condition is absent.	2	C5	4
2.a	Consider the following allocated memory system:	3	C3	6

Hole 35 K	Used 63 K	Used 432 K	Hole 235 K	Used 122 K	Hole 80 K	Used 30 K	Hole 56 K	Used 38 K	Hole 75 K

Requests from processes with 190 K, 40 K, 12 K, 60 K and 8 K are received in ascending order. Now show how each process will be allocated using –

i. First-Fit Algorithm
ii. Best-Fit Algorithm
iii. Worst-Fit Algorithm

Rank the algorithms in terms of how efficiently they use memory.

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2.b	How Hit and Miss occurrences are handled in Translated Look Aside Buffer? Also explain its significance with necessary diagram.	3	C5	4
	Group B			
3.a	Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0 , 1. Assuming demand paging with three frames, what will be the Hit Ratio and Miss Ratio for the following replacement algorithms? i. LRU replacement ii. FIFO replacement iii. Optimal replacement	3	C5	6
3.b	An operating system shows very high page fault rates, low CPU utilization, and excessive disk activity. 1. Identify the problem. 2. Explain why it occurs. 3. Suggest a method to solve it.	3	C5	4
	Or			
3.a	A virtual memory system has 4 page frames. A process generates the page reference string: 7, 2, 5, 1, 2, 7, 3, 4, 2, 1, 5, 6, 2, 7, 4, 3. Using this sequence, compare LRU and Optimal page replacement algorithms. Determine the number of page faults for each, show how the frames are updated step by step, and identify which algorithm performs better with justification.	3	C5	6
3.b	In a computer system, performance degrades and CPU utilization drops as more processes are loaded into memory. Explain thrashing, its impact on system performance, the conditions that cause it, its relation to the working set, and how the operating system can prevent or reduce it.	3	C5	4
4.a	Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 2150 and the previous request was at cylinder 1805. The queue of pending requests in FIFO order is: 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? a. FCFS b. SCAN c. C-SCAN d. LOOK	3	C4	6
4.b	Explain Access matrix with domains as objects and Access matrix with Owner rights containing the special access rights: read, switch, control, owner.	3	C2	4
5.a	Beyond just categorizing threats, how do we distinguish between vulnerabilities that specifically target software applications (program threats) and those that exploit weaknesses in the underlying infrastructure, operating systems, or networks (system threats) and what are some prominent real-world examples of each?	2	C5	6
5.b	How do widely adopted computer security classification schemes help us design, implement and evaluate security systems across the globe?	2	C5	4

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